THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): A cricketer moves his hands backward while catching a fast ball. Reason (R): This increases the time of stopping and hence decreases the force on the hands. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A loaded truck of mass 4000 kg and an empty truck of mass 1000 kg are pushed by equal forces. The ratio of their accelerations (loaded : empty) is: 1

(a) 1 : 1 **(b)** 4 : 1 **(c)** 1 : 4 **(d)** 2 : 1

3. A book is placed on a table. The reaction to the weight of the book is: 1

(a) the normal force by the table **(b)** friction
(c) the push of the table **(d)** the pull of the book on the Earth

4. A 5 kg block on a smooth horizontal floor is pulled by a horizontal force of 20 N. Its acceleration is: 1

(a) 5 m/s squared **(b)** 20 m/s squared **(c)** 2 m/s squared **(d)** 4 m/s squared

5. Assertion (A): Action and reaction forces never cancel each other. Reason (R): They act on different bodies. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A force of 12 N acts on a body of mass 3 kg for 5 s starting from rest. The final speed is: 1

(a) 12 m/s **(b)** 4 m/s **(c)** 60 m/s **(d)** 20 m/s

7. A heavy box is pushed along the floor with a force F. Which of the following would make the push easier? 1

(a) increasing the weight **(b)** reducing the friction by rollers
(c) pushing downward **(d)** using a rough floor

8. A ball of mass 100 g is thrown with a speed of 20 m/s horizontally and hits the wall to stop in 0.01 s. The force is: 1

(a) 200 N **(b)** 2 N **(c)** 20 N **(d)** 2000 N

9. **Assertion (A):** Passengers in a bus jerk forward when the bus stops suddenly. 1
Reason (R): The lower part of the body stops with the bus, while the upper part tends to remain in motion because of inertia.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The SI unit of force is the newton. 1
Reason (R): Momentum is the product of the mass and the velocity of a body.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A body of mass 2 kg moving with 10 m/s is acted on by a force of 4 N opposite to its motion. Find the time to stop and the distance. 2

12. Explain why a person at the surface of Earth does not move towards the Earth with acceleration g but is at rest. 2

13. Why does the tension in a rope used to pull a block at constant velocity equal the friction on the block? 2

14. Two forces of 9 N and 5 N act on a 2 kg body in the same direction and then in opposite directions. Find the accelerations. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A car of mass 1500 kg is moving on a level road at 72 km/h. The driver applies the brakes and the car stops in 5 s. Find the force and the distance. Compare the force with the weight. 3

16. A bullet of mass 10 g moving at 400 m/s enters a wooden block and stops in 0.04 m. Find the average retarding force and the time of penetration. 3

17. A hammer of mass 1 kg moving at 10 m/s strikes a nail and stops in 0.01 s. Find the force and the force on the nail. 3

OR

A force of 30 N acts on a 6 kg body initially at rest for 4 s. Find the final velocity, the change in momentum and the distance covered. 3

18. A girl pushes a box with 40 N and the friction is 40 N. Describe the motion and explain what happens when she pushes with 60 N. 3

OR

State Newton's third law and give three examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A porter at a railway station pushes a loaded trolley of mass 20 kg on a level platform with a horizontal force of 50 N. The platform exerts a friction force of 10 N opposing the motion. Take $g = 10$ m/s^2. 5
(i) Find the weight of the trolley.
(ii) Find the net force on it.
(iii) Find its acceleration.
(iv) If the trolley starts from rest, find its velocity and the distance travelled after 5 s.

20. A gun of mass 4 kg fires a bullet of mass 20 g at 200 m/s. Find the recoil velocity of the gun and name the law involved. 5

OR

An object of mass 5 kg is acted on by two forces: 30 N east and 40 N north. Find the magnitude of the net force, the acceleration and the direction. 5

END OF QUESTION PAPER — Total: 40 Marks